

Describe how parameters of image noise can be estimated. Describe different spatial filters used for image restoration and compare their performance.

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Estimation of Noise Parameters

1. The shape of the histogram identifies the closest PDF match to the noise type present.
2. A small flat region of an image (containing only noise, no structure) is selected.
3. The mean μ and variance σ^2 of that region are computed.
4. These parameters are matched against known noise PDFs to identify the noise type.

Spatial Filters for Image Restoration

A. Mean Filters

Arithmetic Mean Filter: Replaces each pixel with the neighborhood's average intensity to smooth local variations. This reduces random noise but simultaneously blurs sharp edges.

Geometric Mean Filter: Uses the product of intensities. It achieves smoothing similar to the arithmetic mean but tends to **lose less image detail** in the process.

Harmonic Mean Filter: Highly effective for **salt noise** and Gaussian noise, but completely fails for pepper noise.

Contra-harmonic Mean Filter: A versatile filter controlled by order Q . Suited for impulse noise. Use $Q > 0$ for **pepper noise** and $Q < 0$ for **salt noise**.

B. Order-Statistics Filters

Median Filter: Replaces a pixel with the median of its neighborhood. It is the most common tool for reducing **impulse noise** without blurring edges.

Max Filter: Finds the brightest points; best for reducing **pepper noise**.

Min Filter: Finds the darkest points; best for reducing **salt noise**.

Midpoint Filter: Averages the maximum and minimum values. It works best for randomly distributed noise like **Gaussian or uniform noise**.

Alpha-trimmed Mean Filter: Deletes the d/2 lowest and d/2 highest intensity values and averages the rest. It is ideal for images corrupted by **multiple types of noise**.

Comparison of Spatial Filters

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Filter	Best For	Main Weakness
Arithmetic Mean	General smoothing	Blurs sharp edges
Geometric Mean	Preserving image detail	Fails if pixels are zero
Harmonic Mean	Salt and Gaussian noise	Fails on pepper noise
Contraharmonic	Specific salt or pepper noise	Highly sensitive to Q value
Median	Salt-and-pepper noise	Can erase fine details
Max	Reducing pepper noise	Increases salt noise
Min	Reducing salt noise	Increases pepper noise
Alpha-trimmed	Mixed noise environments	More complex to compute